INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Infosys SE Complete Preparation — Arithmetic & Geometric Progressions (AP/GP)

Section: prepinsta infosys SE | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

Infosys SE Complete Preparation — Arithmetic & Geometric Progressions (AP/GP)

Source Portal: PrepInsta Infosys Preparation Suite (`https://prepinsta.com/infosys/aptitude/series-and-progression/`)
Target Role: Infosys Systems Engineer (SE) | **Category:** 01_Mathematical_Ability_Technical_Aptitude
Total Verified Questions Extracted: 10
Quizzes Covered: quiz-1, quiz-2, quiz-3, quiz-4, quiz-5, quiz-6

📌 Section 1: Rules & Core Concepts to Keep in Mind

These rules and fundamental principles govern all Infosys questions on this topic:

- Arithmetic Progression (AP): nth term Tn = a + (n - 1)d; Sum Sn = (n/2) * [2a + (n - 1)d] = (n/2) * (first + last).
- Geometric Progression (GP): nth term Tn = a * r^(n - 1); Sum Sn = a * (r^n - 1) / (r - 1).
- Sum of infinite GP (|r| < 1): S_inf = a / (1 - r).

⚡ Section 2: How to Solve Quickly (Exam Elimination Strategies)

Proven time-saving strategies recommended by PrepInsta for Infosys 2026 examination:

1. **Option Elimination via Unit Digits & Multiples:** Before doing full calculations, inspect the options to see if last digits or divisibility by 3, 7, 9, 11 can eliminate 2 to 3 wrong choices.
2. **Approximation & Magnitude Checks:** When exact arithmetic is tedious, round numbers to neat multiples (e.g., 99 ≈ 100) to find the nearest valid option.
3. **Reverse Option Substitution:** In complex algebraic equations, plug option values directly into the question stem to test validity.

🎯 Section 3: High-Yield Tricks & Shortcuts

Key mathematical formulas, mental-math shortcuts, and speed axioms:

- Sum of first n natural numbers = n(n + 1) / 2.

- Sum of first n odd numbers = n^2.
- Sum of squares of first n natural numbers = n(n + 1)(2n + 1) / 6.
- Sum of cubes of first n natural numbers = [n(n + 1) / 2]^2.

📝 Section 4: Comprehensive PrepInsta Practice Question Bank (All Quizzes)

Contains **10** verbatim questions with options, verified answer keys, and complete derivations:

Question 1 (PrepInsta quiz-1 - Q1)

Question Stem:
Find the missing term in the following series?

0,2,9,?,50,90,147

Options:
- **(A)** `20`
- **(B)** `24`
- **(C)** `27`
- **(D)** `36`

Correct Answer: (B) `24`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Ans: 24.



0 ,2, 9, ?, 50, 90, 147.



0,2,9,24,50,90,147 --&gt;

diff b/w two terms

2-0=2,

9-2=7,

24-9=15,

50-24=26,

90-50=40,

147-90=57,



The new series : 2,7, 15, 26, 40, 57

diff &amp;amp; new series: 5, 8, 11, 14, 17

diff : 3, 3, 3, 3
```

Question 2 (PrepInsta quiz-1 - Q2)

Question Stem:

The aggregate number of whole numbers somewhere around

100 and 200 which are distinct by both 9 and 6 is:

Options:

- **(A)** `5`
- **(B)** `6`
- **(C)** `7`
- **(D)** `8`

Correct Answer: (A) `5`

PrepInsta Detailed Explanation & Mathematical Derivation:

LCM of 6 and 9: 18

>6=2*3

>9=3*3

What is the highest multiple of 18 under 100?

>200-100=100

90 = 18*5

Calculate the first starting position, 108

>108=18*6

If 108 + 90 < 200 AND 200−(108+90) < 18

then 90/18 is the answer.

```
The answer is 5
```

```
 
```

Question 3 (PrepInsta quiz-1 - Q3)

Question Stem:

A few buy National Savings Certificates each year whose worth surpasses the earlier years buy by Rs 400. Following 8 years, she finds that she has obtained declarations whose aggregate face worth is Rs 48000. What is the face estimation of the Certificates acquired by her in the first years?

Options:

- **(A)** `Rs 4300`
- **(B)** `Rs 4400`
- **(C)** `Rs 4500`
- **(D)** `Rs 4600`

Correct Answer: (D) `Rs 4600`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Let the required value be Rs a.

Also d= 400, n = 8 and Sn = 48000.

Sn   = n/2 [2a + (n -1) d] ⇒ 8/2 *[2a + 7 *400] = 48000

⇒ 2a + 2800 = 12000⇒ 2a=9200⇒ a = 4600
```

Question 4 (PrepInsta quiz-1 - Q4)

Question Stem:

Which term of the G.P. 5,10,20,40 ...is 1280?

Options:

- **(A)** `Eighth`

- **(B)** `ninth`
- **(C)** `tenth`
- **(D)** `eleventh`

Correct Answer: (B) `ninth`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Let, it be the n-th term of this GP series.



a = 5; r = 2.



1280 = 5*{2^(n - 1)}



{2^(n - 1)} = 256 = (2^8)



So, (n- 1) = 8; therefore, n = 9.
```

Question 5 (PrepInsta quiz-1 - Q5)

Question Stem:

The total of every single common number from 75 to 97 is:

Options:

- **(A)** `1598`
- **(B)** `1798`
- **(C)** `1958`
- **(D)** `1978`

Correct Answer: (D) `1978`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
The sum of an arithmetic series is the average of the first and last terms, times the number of terms.



For this series, the average of the first and last terms is (97+75)/2=86(97+75)/2=86



For any arithmetic series with a common difference of 1, the number of terms is the largest term minus the smallest term plus 1.



For this series, that is 97−75+1=2397−75+1=23



Therefore, the sum of this arithmetic series is 86⋅23=1978
```

Question 6 (PrepInsta quiz-1 - Q6)

Question Stem:
The number of inhabitants in microbes, society copies ever

2 minutes. How long will it take for the populace to develop

from 1000 to 512000 microbes?

Options:
- **(A)** `10`
- **(B)** `12`
- **(C)** `14`
- **(D)** `18`

Correct Answer: (D) `18`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Let us assume it take n time units to reach 512000 microbes, where each time unit = 2 minutes



Number of microbes in nth time unit=1000*2n=1000*2n



1000*2n=5120001000*2n=512000



n=9n=9



Thus, total time taken = 9*2 = 18 minutes
```

Question 7 (PrepInsta quiz-1 - Q7)

Question Stem:

Discover three numbers in A.P. whose sum is 15 and item is 80.

Options:

- **(A)** `1,4 and 9 or 9,4, and 1`
- **(B)** `3,5 and 9 or 9,5, and 3`
- **(C)** `3,6 and 9 or 9,6, and 3`
- **(D)** `2,5 and 8 or 8,5, and 2`

Correct Answer: (D) `2,5 and 8 or 8,5, and 2`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Let the numbers be (a-d), an and (a+d). At that point,




(a-d) +a+ (a+d) = 15 ⇒ 3a = 15 ⇒ a = 5



(a-d)*a*(a+d) = 80 ⇒ (5-d)*5 * (5+d) = 80



⇒ (25-d2) = 16 = d2 =9 ⇒ d = 3



Numbers are 2, 5, 8 or 8, 5, 2.
```

Question 8 (PrepInsta quiz-1 - Q8)

Question Stem:
The sixth term of an A.P. is 12 and its eighth term is 22.

Locate its first term, normal contrast and sixteenth term.

Options:
- **(A)** `61`
- **(B)** `62`
- **(C)** `63`
- **(D)** `64`

Correct Answer: (B) `62`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Let, first term = a and normal contrast =d.


T6 = 12 ⇒ a+5d= 12 …. (i)


T8= 22 ⇒ a+7d = 22 … (ii)


On subtracting (i) from (ii), we get 2d = 10 ⇒ d = 5




Putting d= 5 in (i), we get a+5*5 = 12 ⇒ a= (12-25) =-13


∴ First term = - 13, normal distinction = 5.




T16= a+ 15d = - 13+15*5 = (75-13) = 62
```

Question 9 (PrepInsta quiz-1 - Q9)

Question Stem:
Find the first term of an AP whose 8th and 12th terms are

respectively 39 and 59.

Options:
- **(A)** `5`
- **(B)** `4`

- **(C)** `3`
- **(D)** `6`
- **(E)** `7`

Correct Answer: (B) `4`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
1st Method:

8th term = a+7d = 39 ..... (i)

12th term = a+11d = 59 ..... (ii)

(i)-(ii);



Or, a+7d-a-11d = 39-59; Or, 4d = 20;

Or, d = 5;

Hence, a+7*5 = 39;

Thus, a = 39-35 = 4.

2nd Method (Thought Process):

8th term = 39;

And, 12th term = 59;

Here, we see that 20 is added to 8th term 39 to get 12th term 59 i.e. 4 times the common
difference is added to 39.

So, CD = 20/4 = 5.

Hence, 7 times CD is added to 1st term to get 39. That means 4 is the 1st term of the AP.
```

Question 10 (PrepInsta quiz-1 - Q10)

Question Stem:
What will be unit's digit in the result of the expression

3^65 * 6^53 *9^5?

Options:
- **(A)** `6`
- **(B)** `2`
- **(C)** `3`
- **(D)** `7`

Correct Answer: (B) `2`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
4.72




unit is 2
```